A Model-Based Approach for Recognizing Unknown Goal Combinations

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Introduction and Background

Plan, activity, and intent recognition (PAIR for short; inclusive of *goal recognition* and other variants is an abductive inference process where an agent generates candidate explanations (recognition hypotheses) as to the unobserved future actions and goals of another agent, based on observations of the observed agent's interactions with its environment (Sukthankar et al. 2014; Mirsky, Keren, and Geib 2021; Meneguzzi and Fraga Pereira 2021).

Current approaches to goal recognition assume the recognizer rely on a domain model (describing actions and their effects), and a set of potential goals G. Given these, a *Model-Based Recognizer (MBR)* (Masters and Vered 2021) can use a planner to dynamically generate candidate plans that match the observed agent's actions. Assuming the observed agent is rational (pursues an optimal plan to its goal), the recognizer can probabilistically rank goals in G as candidate hypotheses (Baker, Tenenbaum, and Saxe 2007; Ramırez and Geffner 2010; Masters and Vered 2021).

An observed agent may follow a plan that combines multiple known goals from G, by interleaving or sequencing actions for their achievement. Given the set G of atomic goals, there are $2^{|G|}$ potential goal combinations (ignoring ordering). Some older library-based methods address this possibility Geib (2004); Avrahami-Zilberbrand, Kaminka, and Zarosim (2005); Hu et al. (2008); Banerjee and Kraemer (2011) without explicit enumeration of the combinations.

However, essentially all work in MBR assumes that the observed agent is pursuing *one* of the known goals in G, and thus cannot address goal combinations unless the combinations are given explicitly. This is clearly infeasible. For instance, MBR methods that rely on a planner would make $\mathcal{O}(2^{|G|})$ planner executions. The only exception of which we are aware is by Dann et al. (2023), where recognition of multiple goals whose actions are interleaved is possible, assumes that the observed agent uses a known action-selection mechanism to schedule actions for different goals while an offline process (extending (Amado, Mirsky, and Meneguzzi 2022)) generates goal-recognition information.

We present a novel critical step towards removing this assumption. We propose a heuristic MBR method that considers potential combination goals when there is evidence that recognition hypotheses with known goals are insufficient to explain the observations under the assumption of rationality. To avoid explicit enumeration of such goals, the heuristic method works *lazily*: it considers only a subset of combination goals, and only when they improve the hypotheses set. Below, we briefly present the heuristic method, and report on very promising results from experiments, demonstrating the efficacy of the approach.

Recognizing Goal Combinations

A *combinatorial* goal recognition problem is a tuple $T = (P, O, \mathcal{G}, \mathcal{C})$, where $P = \langle F, I, A \rangle$ is a planning domain, and $O = \langle o_1, o_2, \ldots o_m \rangle$, is the action sequence that has been observed, $o_i \subseteq A$, where $1 \leq i \leq m$. $\mathcal{G}$ is a set of *atomic* goals, $\mathcal{G} \subseteq 2^F$. $\mathcal{C}$ is a set of *combination operators* that can be used to combine members of $\mathcal{G}$ to create *goal combinations*. The closure of $\mathcal{G}$ with respect to the operators in $\mathcal{C}$ is denoted here G. This is the set of all potential goal combinations that may be described by applying any sequence of operators in C, to atomic goals in $\mathcal{G}$. The classical MBR definition of a goal recognition problem is a triplet $(P, O, \mathcal{G})$, which is a special case, where $\mathcal{C}$ contains only the identity operator $I(g_1) = g_1$, and therefore $G = \mathcal{G}$.

A solution to T is a *subset* of $R \subseteq G$ such that for all $g \in R$, the probability $P(g|O) > 0$, and for all $\underline{g} \notin R$, the probability $P(\underline{g}|O) = 0$. A solution is considered *correct* if given a ground-truth goal $g^* \in G$, $g^* = \arg\max_{g \in R} P(g|O)$, i.e., the probability of the ground-truth goal g^* is maximal in R.

Our focus is on combination goal recognition problems, where the closure set G is not taken to be identical to $\mathcal{G}$ as in classical MBR definitions, but instead, $\mathcal{C}$ contains both the identity operator, as well as the set union operator. Then, $G \subseteq 2^{\mathcal{G}}$ (the powerset of $\mathcal{G}$), i.e., any *unordered* combination of atomic goals in $\mathcal{G}$.

Existing methods construct R by explicitly considering every goal in G. One could think of these as employing a *generate and test* strategy: For every candidate $g \in G$, compute $P(g|O)$. If it is greater than 0, add to R. While steady progress has been made on heuristically alleviating the computational complexity and practical runtime of this process (Masters and Sardina 2017; Pereira, Oren, and Meneguzzi 2017; Vered and Kaminka 2017; Vered et al. 2018; Masters and Sardina 2019), it is clear that this strategy does not scale in general.

We take an alternative approach. Rather than testing all potential goals in G, we construct the solution set R incrementally. The OGRE algorithm (Occam's Goal REcognizer)

begins by setting $R_0 = \mathcal{G}$, i.e., the atomic goals form the basis for R. With each observation o_i ($i < n$), in order, it proceeds to compute a ranking score $s(g|O_i)$, where $g \in R$ and O_i is the (sub)sequence of observations $\langle o_1, \ldots o_i$. The computation can use any standard MBR method; in the experiments below we use the definition of Kaminka, Vered, and Agmon (2018). Typically, $s(g|O_i)$ is used to compute $P(g|O_i)$ by normalization, and any goal $g \in R_{i-1}$ for which $P(g|O_i) = 0$ is pruned from R_i.

However, OGRE differs from previous approaches in this step. Instead of simply looping back to consider the next observation, it looks for changes in $s(g|\cdot)$, resulting from extending the observation sequence by the last observation o_i. It treats these changes as evidence that perhaps additional goal combinations merit explicit consideration in R_i. In other words, OGRE is a lazy-computation algorithm: it only adds to R as necessary, and does not generate candidates in R_i unless they are needed. To do this, OGRE has two key decision steps.

Deciding to Add Goals to R_i. OGRE adds new combined goals to the hypotheses (solution) set R when there is evidence that none of the goals currently considered is the agent's real (hidden) goal. OGRE uses a threshold θ for $s(\cdot)$ to determine whether a goal's score is within a *bounded rationality distance* from the perfect score, which would indicate that the observations perfectly match the optimal plan that would have been followed by an optimal agent pursuing g. If *no* goal $g \in R_i$ satisfies $s(g|O_i) \geq \theta$, OGRE considers this an indication that the candidate set R_i is insufficient. A threshold θ may needed in practice, as $s(\cdot)$ can be inherently noisy (it is an estimate, after all), or it may be that the observed agent generating O_i is not perfectly rational (e.g., because it has bounded resources). Nevertheless, in the experiments, we set $\theta = 1$, as we experiment with sequences generated by an optimal planner.

We use the goal mirroring scoring function $s(g|O_i) := \frac{|\pi_g|}{|m^-_{i,g}|+|m^+_{i,g}|}$ due to Vered and Kaminka (2017), where: π_g is an optimal plan from the initial state to the goal g, $m^-_{i,g}$ is the observations seen so far, and $m^+_{i,g}$ is an optimal plan from the current state of the agent (after the the execution of the observed actions), to the goal g. $s() = 1$ when the agent's observed actions are perfectly optimal with respect to π_g, an optimal plan to reach g (ignoring the observations).

Choosing Goals to Add to R_i. To decide on candidate additions to R_i, OGRE maintains a *contribution table* $M(g, o)$. For every observation $o_1, \ldots o_i$, and for every goal $g \in \mathcal{G}$, $M(g, o)$ marks whether o *contributes* to g (a true/false value). The observation o is considered to contribute towards a goal g, if the plan $m^+_{j,g}$ is *shorter* after the agent performs the observed action o_j, for any $j \leq i$ (intuitively, the action took the agent a step closer to the goal). Formally,

$$M(g, o_j) = \top \iff |m^+_{j-1,g}| > |m^+_{j,g}|. \quad (1)$$

Note that M is built incrementally with every step, so its construction is not computationally burdensome.

When the algorithm decides to add new goals after observation o_i is received, M is first updated. It is then scanned to construct subsets of size K, $H \subseteq \mathcal{G}$, $|H| = K$, such that the members of H are sufficient to have a true value in all observations. In other words, the subset H is the minimal set of goals whose $M()$ values, taken with a logical *or* over all observations $o_1, \ldots o_i$, result in $\top$.

As constructing H is intractable, do this incrementally. First, we begin with $K = 2$, and anchor H with goals in R_{i-1} whose contribution count are sufficient given the size of the combination goal. Thus when OGRE considers a two-goal combination, it first anchors H with goals h where $|\{M(h, \cdot)\}|$ is at least half of the number of observations $|O_i|$. Generally, we add h to as an anchor to a K-combination when $\frac{|\{M(h,\cdot)\}|}{|O_i|} \geq \frac{1}{K}$. Once the anchor set is known, we search for additional goals h to add to it (possibly other anchors, but not limited to them), to construct a combination H. We add to R_i any combination H that meets the criterion $P(H|O_i) \geq \theta$ (i.e. any combination of size K). If at least one such combination is found, the construction of R_i stops: it is now R_{i-1}, minus any goals that can be pruned, and with additional combinations H of size K. If no combination H is found that is within the threshold θ from the perfect score, we increase K and repeat the search for H. When the loop iterating over K terminates, either R_i necessarily has goal combinations that should be considered, or no such combinations have been found.

Experiments and Conclusions

To evaluate the heuristic we presented, we performed 38 experiments in known domains (*ferry*, *driverlog*), used previously in plan recognition benchmarks[1]. The experiments were constructed such that existing known atomic goals used in previous goal recognition research were used as the atomic goal set $\mathcal{G}$. In addition, a single random 2- or 3-goal combination was constructed from these, to serve as the true goal driving the generated observation sequence, on which OGRE was deployed. It was given $\mathcal{G}$ but not the true goal.

Results. In 37% of the experiments, OGRE gave the correct goal the highest likelihood. In additional 58%, OGRE did not generate the combined goal, as the observations were consistent with a shorter combination; i.e., OGRE did not generate the correct K-combination goal, because it generated a simpler goal hypothesis that is consistent with the observations. In the remaining 5% of the experiments, OGRE gave the highest score for more than one generated K-combination goal, including the correct one.

Conclusions. We consider these results to be very promising. OGRE heuristically recognized goal combinations not known to it a-priori, without considering increasing the size of the original hypothesis set exponentially. The results reveal that generating synthetic problems is challenging, as simpler explanations were unintentionally made possible. We look forward to examining this and other challenges.

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[1] https://github.com/ramonpereira/OnlineGoalRecognition-DiscreteDomains

References

Amado, L.; Mirsky, R.; and Meneguzzi, F. 2022. Goal Recognition as Reinforcement Learning. *Proceedings of the AAAI Conference on Artificial Intelligence*, 36(9): 9644–9651.

Avrahami-Zilberbrand, D.; Kaminka, G. A.; and Zarosim, H. 2005. Fast and complete plan recognition: Allowing for duration, interleaved execution, and lossy observations. In *IJCAI workshop on Modeling Other agents from Observations (MOO-05)*. Scotland, Edinburgh.

Baker, C. L.; Tenenbaum, J. B.; and Saxe, R. R. 2007. Goal inference as inverse planning. In *Proceedings of the 29th Annual Meeting of the Cognitive Science Society*.

Banerjee, B.; and Kraemer, L. 2011. Branch and Price for Multi-Agent Plan Recognition. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 601–607.

Dann, M.; Yao, Y.; Alechina, N.; Logan, B.; Meneguzzi, F.; and Thangarajah, J. 2023. Multi-Agent Intention Recognition and Progression. In *Proceedings of the International Joint Conference on Artificial Intelligence*, 91–99.

Geib, C. W. 2004. Assessing the Complexity of Plan Recognition. In *Proceedings of the Nineteenth National Conference on Artificial Intelligence (AAAI-04)*.

Hu, D. H.; Pan, S. J.; Zheng, V. W.; Liu, N. N.; and Yang, Q. 2008. Real World Activity Recognition with Multiple Goals. In *Proceedings of Ubiquitous Computing (UbiComp-08*.

Kaminka, G. A.; Vered, M.; and Agmon, N. 2018. Plan Recognition in Continuous Domains. In *Proceedings of the AAAI Conference on Artificial Intelligence*.

Masters, P.; and Sardina, S. 2017. Cost-Based Goal Recognition for Path-Planning. In *Proceedings of the International Joint Conference on Autonomous Agents and Multi-Agent Systems*, 750–758. International Foundation for Autonomous Agents and Multiagent Systems.

Masters, P.; and Sardina, S. 2019. Cost-Based Goal Recognition in Navigational Domains. *Journal of Artificial Intelligence Research*, 64: 197–242.

Masters, P.; and Vered, M. 2021. What's the Context? Implicit and Explicit Assumptions in Model-Based Goal Recognition. In *Proceedings of the International Joint Conference on Artificial Intelligence*, 4516–4523.

Meneguzzi, F.; and Fraga Pereira, R. 2021. A Survey on Goal Recognition as Planning. In *Proceedings of the Thirtieth International Joint Conference on Artificial Intelligence*, 4524–4532. Montreal, Canada: International Joint Conferences on Artificial Intelligence Organization. ISBN 978-0-9992411-9-6.

Mirsky, R.; Keren, S.; and Geib, C. 2021. Introduction to Symbolic Plan and Goal Recognition. *Synthesis Lectures on Artificial Intelligence and Machine Learning*, 15(1): 1–190. Publisher: Morgan & Claypool Publishers.

Pereira, R. F.; Oren, N.; and Meneguzzi, F. 2017. Landmark-Based Heuristics for Goal Recognition. In *Proceedings of the AAAI Conference on Artificial Intelligence*.

Ramırez, M.; and Geffner, H. 2010. Probabilistic plan recognition using off-the-shelf classical planners. In *International Joint Conference on Artificial Intelligence*.

Sukthankar, G.; Goldman, R. P.; Geib, C.; Pynadath, D. V.; and Bui, H., eds. 2014. *Plan, Activity, and Intent Recognition*. Morgan Kaufmann.

Vered, M.; and Kaminka, G. A. 2017. Heuristic Online Goal Recognition in Continuous Domains. In *Proceedings of the International Joint Conference on Artificial Intelligence*, 4447–4454. An improved version (with minor corrections) is available as arxiv:1709.09839.

Vered, M.; Pereira, R. F.; Magnaguagno, M. C.; Meneguzzi, F.; and Kaminka, G. A. 2018. Online Goal Recognition as Reasoning over Landmarks. In *AAAI workshop on Plan-, Activity-, and Intent- Recognition (PAIR)*.